

Ex. NO: 1

Count Uppercase Letters

PROGRAM:

```
import java.util.Scanner;

public class CountUppercaseLetters {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        int count = 0;

        for (int i = 0; i < str.length(); i++) {

            char ch = str.charAt(i);

            if (Character.isUpperCase(ch)) {

                count++;

            }

        }

        System.out.println(count);

    }

}
```

Ex. No: 2

Count LowerCase Letters

PROGRAM

```
import java.util.Scanner;

public class CountLowercaseLetters {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        int count = 0;

        for (int i = 0; i < str.length(); i++) {

            char ch = str.charAt(i);

            if (ch >= 'a' && ch <= 'z') {

                count++;

            }

        }

        System.out.println(count);

    }

}
```

Ex. No: 3

Count Digits

PROGRAM:

```
import java.util.Scanner;

public class CountDigits {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        int count = 0;

        for (int i = 0; i < str.length(); i++) {

            char ch = str.charAt(i);

            if (ch >= '0' && ch <= '9') {

                count++;

            }

        }

        System.out.println(count);

    }

}
```

Ex. No: 4

Count Special Characters

PROGRAM:

```
import java.util.Scanner;

public class CountSpecialCharacters {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        int count = 0;

        for (char ch : str.toCharArray()) {

            if (!Character.isLetterOrDigit(ch) && ch != ' ') {

                count++;

            }

        }

        System.out.println(count);

    }

}
```

Ex. No: 5

Remove All Spaces

PROGRAM:

```
import java.util.Scanner;

public class RemoveAllSpaces {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        System.out.println(str.replace(" ", ""));

    }

}
```

Ex. No: 6

Find the Frequency of a Character

PROGRAM

```
import java.util.Scanner;

public class FrequencyCharacter {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        char ch = sc.next().charAt(0);

        int count = 0;

        for (int i = 0; i < str.length(); i++) {

            if (str.charAt(i) == ch) {

                count++;

            }

        }

        System.out.println(count);

    }

}
```

Ex. No: 7

Find Duplicate Characters

PROGRAM:

```
import java.util.LinkedHashMap;
import java.util.Map;
import java.util.Scanner;

public class DuplicateCharacter {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        LinkedHashMap<Character, Integer> map = new LinkedHashMap<>();

        for (char ch : str.toCharArray()) {

            map.put(ch, map.getOrDefault(ch, 0) + 1);

        }

        for (Map.Entry<Character, Integer> e : map.entrySet()) {

            if (e.getValue() > 1) {

                System.out.println(e.getKey());

            }

        }

    }

}
```

Ex. No: 8

Check Anagram

PROGRAM:

```
import java.util.Arrays;
import java.util.Scanner;

public class CheckAnagram {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String s1 = sc.nextLine();

        String s2 = sc.nextLine();

        char[] a = s1.toCharArray();

        char[] b = s2.toCharArray();

        Arrays.sort(a);

        Arrays.sort(b);

        if (Arrays.equals(a, b))

            System.out.println("Anagram");

        else

            System.out.println("Not Anagram");

    }

}
```

Ex. No: 9

Toggle Case

PROGRAM:

```
import java.util.Scanner;

public class ToggleCase{

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String result = " ";

        for (char ch : str.toCharArray()) {

            if (Character.isUpperCase(ch))

                result += Character.toLowerCase(ch);

            else if (Character.isLowerCase(ch))

                result += Character.toUpperCase(ch);

            else

                result += ch;

        }

        System.out.println(result);

    }

}
```

Ex. No: 10

Find the Longest Word

PROGRAM:

```
import java.util.Scanner;

public class LongestWord {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String[] words = str.split(" ");

        String longest = words[0];

        for (String word : words) {

            if (word.length() > longest.length())

                longest = word;

        }

        System.out.println(longest);

    }

}
```

Ex. No: 11

Find the Shortest Word

PROGRAM:

```
import java.util.Scanner;

public class ShortestWord {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String[] words = str.split(" ");

        String shortest = words[0];

        for (String word : words) {

            if (word.length() < shortest.length())

                shortest = word;

        }

        System.out.println(shortest);

    }

}
```

Ex. No: 12**Reverse Each Word****PROGRAM:**

```
import java.util.Scanner;

public class ReverseEachWord {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String[] words = str.split(" ");

        for (String word : words) {

            for (int i = word.length() - 1; i >= 0; i--) {

                System.out.print(word.charAt(i));

            }

            System.out.print(" ");

        }

    }

}
```

Ex. No: 13**Sort Characters Alphabetically****PROGRAM:**

```
import java.util.Arrays;
import java.util.Scanner;

public class SortCharactersAlphabetically {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String str = sc.nextLine();
        char[] arr = str.toCharArray();
        Arrays.sort(arr);
        System.out.println(new String(arr));
    }
}
```

Ex. No: 14**Print Each Character with ASCII Value****PROGRAM:**

```
import java.util.Scanner;

public class CharacterASCIIValue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String str = sc.nextLine();
        for (char ch : str.toCharArray()) {
            System.out.println(ch + " " + (int) ch);
        }
    }
}
```

Ex. No: 15**Compress a String****PROGRAM:**

```
import java.util.Scanner;

public class ConcatenateTwoStrings {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String str = sc.nextLine();

        String result = " ";

        int count = 1;

        for (int i = 0; i < str.length() - 1; i++) {

            if (str.charAt(i) == str.charAt(i + 1))

                count++;

            else {

                result += str.charAt(i);

                result += count;

                count = 1;

            }

        }

        result += str.charAt(str.length() - 1);

        result += count;

        System.out.println(result);

    }

}
```